

Sonali Shaw

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EDUCATION

University of Chicago

Expected Graduation 2029

Major GPA: 4.0 / 4.0; GPA: 3.9 / 4.0

Major in Computer Science with an intended specialization in Machine Learning with a minor in Chemistry

Relevant Coursework: Python programming, systems programming, proof-based discrete mathematics, linear algebra, multivariable calculus, differential equations, honors chemistry, biology, humanities

University of Chicago Computational Life Sciences Cohort

September 2025–June 2026

Accepted into a highly selective cohort for students to learn, research, and explore careers at the intersection of computer science and biology

University of Chicago Laboratory Schools

September 2021–June 2025

GPA: 3.98 / 4.0; Relevant courses: Linear Algebra & Multivariable Calculus, AP Calculus BC, Discrete Math & Statistics, AP Computer Science, AT Physics, AT Biology, AT Accelerated Chemistry

Inspirit AI: Introduction to Artificial Intelligence and Machine Learning

December 2022–January 2023

Developed several versions of a Python program that analyzed the sentiment of tweets to predict stock market changes

Relevant topics: linear regression, logistic regression, NLP, neural networks, and CNNs

Harvard Summer Session: Molecular Oncology

July 2023–August 2023

Residential summer program covering cancer biology and wet lab techniques used for research

TECHNICAL EXPERIENCE

University of Chicago Data For Common Good

Software Engineer, June 2026–August 2026

- Engineered backend data infrastructure for a clinical research platform, from schema-driven tooling to synthetic data generation
- Debugged and resolved complex full-stack issues in a production-grade clinical trial matching web application

Argonne National Labs & University of Chicago Data Science Institute

Research Assistant, June 2024–August 2024

- Designed and fine-tuned CNNs to analyze aerial bovine imagery and predict bovine body composition
- Utilized several different technical approaches (classification/regression), finally achieving 94–97% accuracy
- Participated in several presentations and a final research symposium

Global IP Law Group

Software Developer Intern, June 2023–July 2023

- Designed and implemented a tool to automate claim chart generation given a patent number using web crawling and web scraping
- Utilized NLP for semantic analysis of claim chart language
- Deployed tool as a user-friendly website integrated with AWS

Director of Student Technology Services at the UChicago Lab Schools (Student Council, Executive Board)

September 2023–June 2025

- Developed and augmented the student council website using React and Firebase (~700 users)
- Integrated the Borda voting system for all-school student council elections
- Served on the student council executive board, organized events, and participated in bills

Standard ML Tutor

Teacher's Assistant, September 2024–June 2025

- Assisted in teaching functional programming for a younger student (age 10) while concurrently learning the material

SCUBA: Summer Coding at UChicago: Builder's Academy (UChicago Hillel Center)

Summer Intern, June 2025–July 2025

- Assisted in developing a written material (teacher's manual and exercises) for teaching high school students introductory python
- Assisted in managing a large fundraising database

O2M Technologies

Research Assistant, May 2025–June 2025

- Assisted in the purification of OX071 and the synthesis of HOPE71

Advanced Chemistry (AT Accelerated Chemistry)

Teacher's Assistant, September 2024–June 2025

- Prepared and assisted with labs, guided students in developing procedures, assisted with written lab reports, provided study help

LEADERSHIP AND AWARDS

American Computer Science League Finals Qualifier (International level)

Science Olympiad: 1st place in Bridge, Cell Biology, Code Busters, and Fermi Questions at Regionals (Illinois, 2023)

Captain of Varsity Dance Team

Captain, October 2023–June 2025

SKILLS

Python, C, Java, JavaScript, Standard ML, PyTorch, NumPy, pandas, scikit-learn, Matplotlib, React, Node.js, Firebase, Git/GitHub, AWS, CNNs, NLP, regression & classification modeling, web scraping & crawling